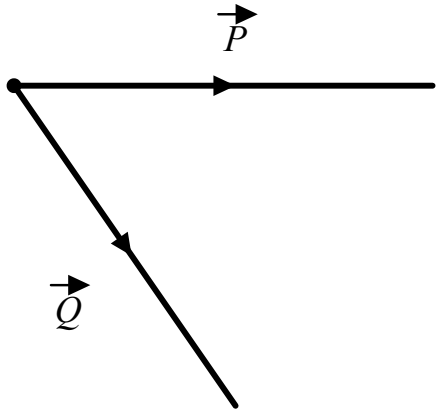
STPM 2020 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ m\ s^{-2}$ and the values of constants printed in the question paper.

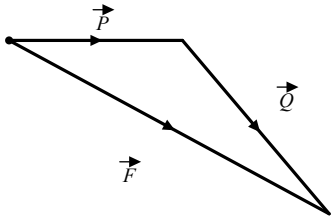
Section A [15 marks]

1. Two forces, ***P*** and ***Q***, act on a point as shown in the diagram below.

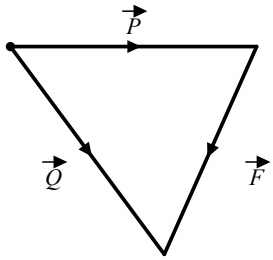


Which vector diagram shows the correct resultant force, ***F***, acting on the point?

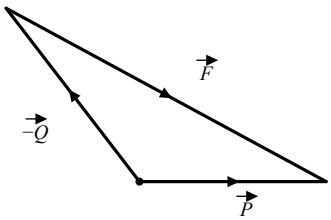
A



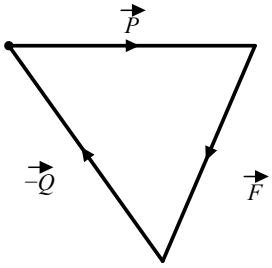
B



C



D



Answer: A

Place the tail of ***Q*** at the head of ***P*** (tip-to-tail). The resultant ***F*** is drawn from the tail of ***P*** to the head of ***Q***.

Explanation: In the triangle of vectors the two forces follow each other head to tail, and the resultant closes the triangle from the starting point to the finishing point. In B and D the arrows do not follow each other correctly, and C uses $-\boldsymbol{Q}$, which gives $\boldsymbol{P} - \boldsymbol{Q}$.

2. A boy releases a small stone from rest at a height of 15 m from the ground. After 1.0 s, he throws another small stone straight down with an initial speed. If the two stones reach the ground at the same time, what is the initial speed of the second stone?

A $10\ \text{m s}^{-1}$ B $13\ \text{m s}^{-1}$ C $16\ \text{m s}^{-1}$ D $30\ \text{m s}^{-1}$

Answer: C

$$\text{First stone: } 15 = \frac{1}{2}(9.81)t^2 \Longrightarrow t = 1.749\ \text{s}$$

$$\text{Second stone falls for } 1.749 - 1.0 = 0.749\ \text{s}$$

$$15 = u(0.749) + \frac{1}{2}(9.81)(0.749)^2 \Longrightarrow u = 16\ \text{m s}^{-1}$$

Explanation: Both stones fall the same 15 m, but the second stone has 1.0 s less time, so it must be thrown down with an initial speed. Find the time of the first stone, subtract 1.0 s, then apply $s = ut + \frac{1}{2}at^2$ to the second stone.

3. The particle P moves with constant velocity and collides head-on with particle Q which is stationary. After the collision, both particles move with the same velocity. Which statement is true about the collision?

A The increase in momentum of particle Q is equal to the decrease in momentum of particle P.
B The increase in kinetic energy of particle Q is equal to the decrease in kinetic energy of particle P.
C The total kinetic energy after the collision of particles P and Q is bigger than the total kinetic energy before the collision.
D The force acted by particle Q on particle P is less than the force acted by particle P on particle Q during the collision.

Answer: A

Momentum is conserved in every collision, so the momentum gained by Q equals the momentum lost by P.

Explanation: The particles move together afterwards, so this is a perfectly inelastic collision: kinetic energy is lost (B and C are wrong). By Newton's third law the forces the particles exert on each other are equal and opposite (D is wrong).

4. An object of mass 0.35 kg is pulled from rest along the rough surface of a table by a horizontal force of 5.0 N. If the velocity of the object is $4.0\ \text{m s}^{-1}$ after it moves 1.2 m, what is the coefficient of friction between the object and the surface of the table?

A 0.05 B 0.78 C 1.41 D 2.14

Answer: B

$$v^2 = 2as \Longrightarrow a = \frac{4.0^2}{2(1.2)} = 6.67\ \text{m s}^{-2}$$

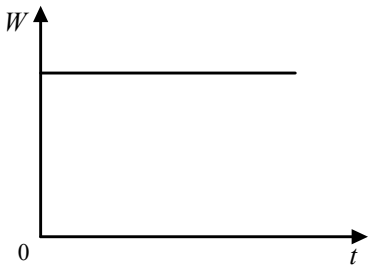
$$F - \mu mg = ma\text{: } 5.0 - \mu(0.35)(9.81) = (0.35)(6.67)$$

$$\mu = 0.78$$

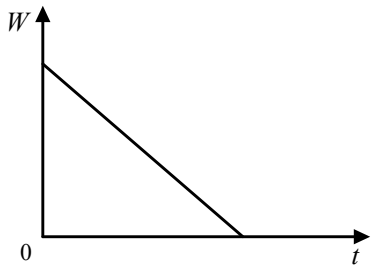
Explanation: Find the acceleration from the motion, then apply Newton's second law with the resultant force = applied force − friction. (The work–energy method gives the same: $5.0 \times 1.2 = \frac{1}{2}(0.35)(4.0)^2 + \mu(0.35)(9.81)(1.2)$.)

5. A body moves in a straight line from rest on a smooth horizontal surface when a constant force acts on it. Which graph shows the variation of work done, W, with time, t, for the motion?

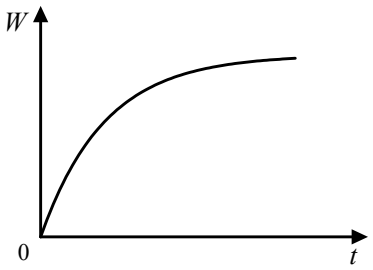
A



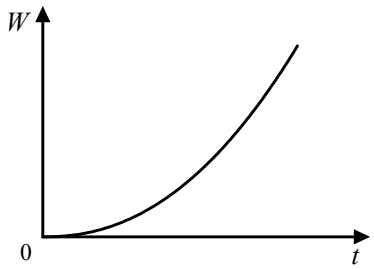
B



C



D



Answer: D

$$W = Fs \text{ and } s = \frac{1}{2}at^2 \Longrightarrow W = \frac{1}{2}Fat^2$$

$W \propto t^2$: a parabola through the origin that curves upwards.

Explanation: The force is constant, but the body speeds up, so it covers more distance in each second. The work done therefore increases faster and faster with time.

6. Two objects, P and Q, perform circular motions at different constant speeds with radius R and $2R$ respectively. The mass of P is $2m$ while that of Q is $3m$. If the centripetal forces of both objects are the same, what is the ratio of speed of P to that of Q?

A $\frac{3}{8}$ B $\frac{\sqrt{3}}{2}$ C $\frac{2\sqrt{3}}{3}$ D $\sqrt{3}$

Answer: B

$$\frac{(2m)v_P^2}{R} = \frac{(3m)v_Q^2}{2R} \Longrightarrow \frac{v_P^2}{v_Q^2} = \frac{3}{4}$$

$$\frac{v_P}{v_Q} = \frac{\sqrt{3}}{2}$$

Explanation: Equate the centripetal forces mv^2/r with the correct mass and radius for each object, and remember to take the square root at the end.

7. A satellite is said to be geosynchronous if it appears to remain over a fixed position on the Earth. Which statement is **not** true about the geosynchronous satellite?

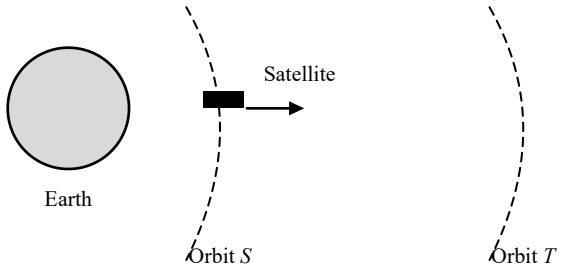
A The satellite completes its orbit in 8.6×10^4 s.
B The satellite can be placed in any chosen orbit depending on its size.
C The satellite moves around the Earth in the same direction as the rotation of the Earth about its axis.
D The satellite moves around the Earth in the circular orbit under the influence of the gravitational force.

Answer: B

A geosynchronous satellite must be at one particular orbital radius (about 4.2×10^7 m from the centre of the Earth), above the equator. Its orbit does not depend on its size or mass, so statement B is not true.

Explanation: Its period equals the Earth's rotation period (8.6×10^4 s, A), it moves in the same direction as the Earth rotates (C), and gravity provides the centripetal force (D). From $T^2 = \frac{4\pi^2 r^3}{GM}$, a fixed period fixes the radius.

8. A satellite is moved from orbit *S* to orbit *T* as shown in the diagram below.



Which is true about the energy of the satellite when it moves from orbits *S* to *T*?

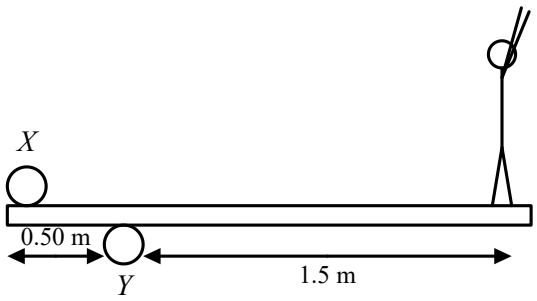
A An increase in potential energy because work is done against gravity.
B A loss in potential energy because work is done against gravity.
C An increase in potential energy because work is done by the gravitational field.
D A loss in potential energy because work is done by the gravitational field.

Answer: A

Orbit *T* is further from the Earth. Moving the satellite outwards requires work to be done against gravity, so its gravitational potential energy increases (becomes less negative).

Explanation: $U = -\frac{GMm}{r}$: as r increases, U increases towards zero. The gravitational field pulls inwards, so it does negative work when the satellite moves outwards.

9. A diver of weight 500 N stands on the free end of a uniform spring board of length 2.0 m as shown in the diagram below.



Two supports, X and Y, which are separated at a distance of 0.50 m to enable the system to be in static equilibrium. If the weight of the spring board is 200 N, what are the magnitudes and directions of the reaction forces acts on the board by X and Y?

	X: Magnitude	*X: Direction*	*Y: Magnitude*	*Y: Direction*
A	700 N	Upward	1400 N	Downward
B	700 N	Downward	1400 N	Upward
C	1700 N	Upward	2400 N	Downward
D	1700 N	Downward	2400 N	Upward

Answer: D

Torques about Y (board weight acts at 1.0 m from X, i.e. 0.50 m beyond Y; diver 1.5 m beyond Y):

$$F_X(0.50) = 500(1.5) + 200(0.50) \Longrightarrow F_X = 1700 \text{ N, downward}$$

$$F_Y = 1700 + 500 + 200 = 2400 \text{ N, upward}$$

Explanation: Both weights try to rotate the board clockwise about Y, so X must pull the board downward to balance them. The vertical forces then balance: the upward force at Y supports all three downward forces.

10. A uniform metal rod of length L, a cross-sectional area A and the Young's modulus E are acted by a tensile force F. The strain energy stored in the rod is represented by

A $\frac{F^2}{EA^2}$ B $\frac{F^2L}{EA}$ C $\frac{F^2}{2EA^2}$ D $\frac{F^2L}{2EA}$

Answer: D

$$U = \frac{1}{2}Fe \text{ and } e = \frac{FL}{EA}$$

$$U = \frac{F^2L}{2EA}$$

Explanation: The strain energy is the area under the force–extension graph, $\frac{1}{2}Fe$. Use the definition of Young's modulus, $E = \frac{FL}{Ae}$, to replace the extension.

11. A container contains an ideal gas at temperature 12.0 °C and pressure 1.00×10^5 Pa. What is the mass of a mole of gas if the density of the gas is 1.20 kg m^{-3}?

A 1.20 g B 3.42 g C 28.4 g D 35.2 g

Answer: C

$$pV = \frac{m}{M}RT \Longrightarrow M = \frac{\rho RT}{p} = \frac{(1.20)(8.31)(285)}{1.00 \times 10^5}$$

$$M = 0.0284 \text{ kg mol}^{-1} = 28.4 \text{ g}$$

Explanation: Write the ideal gas equation in terms of density, $\rho = m/V$. Remember to convert 12.0 °C to 285 K; the answer is in kg mol^{-1} and must be converted to grams.

12. A fixed mass of an ideal diatomic gas is at room temperature, T. Which statement is **not** true about the gas if the mass of a gas molecule is m?

A The kinetic energy per mole of the gas is $\frac{5}{2}RT$.

B The rotational kinetic energy of a molecule of gas is kT.

C The number of degrees of freedom of a molecule of gas is 5.

D The root mean square speed of the molecules of gas is $\sqrt{(\frac{5kT}{m})}$.

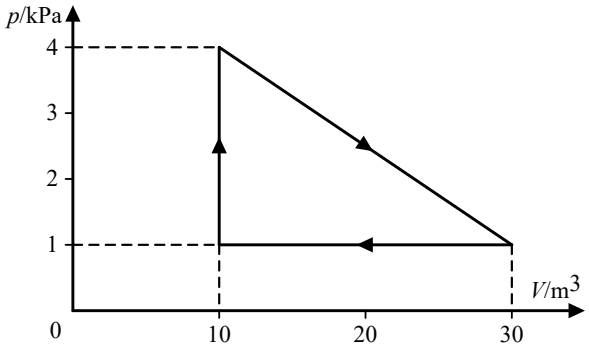
Answer: D

$$c_{\text{r.m.s.}} = \sqrt{(\frac{3kT}{m})}, \text{ not } \sqrt{(\frac{5kT}{m})}$$

So statement D is not true.

Explanation: The r.m.s. speed depends only on the translational kinetic energy, $\frac{3}{2}kT = \frac{1}{2}m\langle c^2\rangle$, whatever the number of degrees of freedom. A diatomic gas has 5 degrees of freedom, total kinetic energy $\frac{5}{2}RT$ per mole, and two rotational degrees giving $2 \times \frac{1}{2}kT = kT$.

13. The cyclic process for an ideal gas is shown in p-V diagram below.



What is net work done by the gas in one cycle?

A 1.0×10^4 J B 2.0×10^4 J C 3.0×10^4 J D 5.0×10^4 J

Answer: C

$$\text{Net work} = \text{area enclosed} = \frac{1}{2}(30 - 10)(4 - 1) \times 10^3 = 3.0 \times 10^4 \text{ J}$$

Explanation: The net work done in a cycle is the area enclosed by the loop. The cycle is clockwise, so the net work is done **by** the gas. Remember that the pressure is in kPa.

14. An ideal gas is heated at constant pressure of 1.0×10^5 Pa. If the increase in the volume of the gas is 400 cm^3 and the increase in the internal energy is 100 J, what is the heat absorbed by the gas?

A 40 J B 60 J C 100 J D 140 J

Answer: D

$$W = p\Delta V = (1.0 \times 10^5)(400 \times 10^{-6}) = 40 \text{ J}$$
$$Q = \Delta U + W = 100 + 40 = 140 \text{ J}$$

Explanation: At constant pressure, the heat absorbed both raises the internal energy and does work as the gas expands. Convert 400 cm^3 to 4.00×10^{-4} m^3.

15. Which statement is **not** true about the transfer of heat by radiation?

A The transfer of heat can occur in vacuum.
B The transfer of heat is via electromagnetic waves.
C The transfer of heat involves the movement of matter.
D The transfer of heat is greater when the object is hotter.

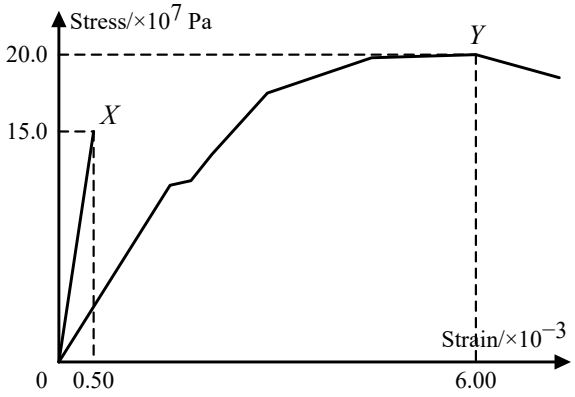
Answer: C

Radiation is transfer of energy by electromagnetic waves; it does not involve the movement of matter. Statement C is not true.

Explanation: Movement of matter is convection. Radiation can travel through a vacuum and the power radiated increases strongly with temperature ($\propto T^4$).

Section B [15 marks]

16. The variation of stress with strain for two different materials, *X* and *Y*, until the breaking point of the materials is shown in the graph below.



(a) Based on the graph, state **two** differences in the properties of *X* and *Y*. Give your reasons. [4]

Answer

X is stiffer than *Y* / *X* has a greater Young's modulus	1
because the gradient of the linear part of *X* is greater	1
X is brittle while *Y* is ductile	1
because *X* breaks at the end of its linear region with no plastic deformation, while *Y* undergoes a large plastic deformation before breaking	1
Alternative difference	
Y has a greater tensile strength (20.0×10^7 Pa) than *X* (15.0×10^7 Pa), because *Y* reaches a higher maximum stress	2
	4

Explanation: Each difference must compare **both** materials and be supported by a reason taken from the graph: the gradient (stiffness), the shape after the proportional limit (brittle or ductile), or the maximum stress (strength).

(b) Determine the Young's modulus of *X*. [2]

Answer

$E = \frac{\text{stress}}{\text{strain}} = \frac{15.0 \times 10^7}{0.50 \times 10^{-3}}$	1
$E = 3.0 \times 10^{11}$ Pa	1
	2

Explanation: Young's modulus is the gradient of the straight-line part of the graph. Remember the factors 10^7 on the stress axis and 10^{-3} on the strain axis.

(c) Determine the required tensile force to break Y if the cross-sectional area is 7.85×10^{-7} m^2. [2]

Answer

$F = \sigma A = (20.0 \times 10^7)(7.85 \times 10^{-7})$ 1

$F = 157$ N 1

2

Total for question **8**

Explanation: The force that breaks Y is found from the maximum (ultimate tensile) stress that Y can withstand, 20.0×10^7 Pa, multiplied by the cross-sectional area.

17. An ideal gas is cooled from 50 °C to 10 °C at constant pressure of 3.00×10^5 Pa. The total heat liberated by the gas is 2.50×10^4 J. The molar heat capacity of the gas at constant volume is 20.76 J mol^{-1} K^{-1} and the molar heat capacity of the gas at constant pressure is 29.07 J mol^{-1} K^{-1}.

(a) Determine the change in internal energy of the gas. [3]

Answer

$Q = nC_{p,\mathrm{m}}\Delta T \Longrightarrow n = \dfrac{-2.50 \times 10^4}{(29.07)(10 - 50)} = 21.5$ mol 1

$\Delta U = nC_{V,\mathrm{m}}\Delta T = (21.5)(20.76)(-40)$ 1

$\Delta U = -1.79 \times 10^4$ J (internal energy decreases by 1.79×10^4 J) 1

3

Explanation: The heat at constant pressure gives the number of moles. The change in internal energy of an ideal gas always uses C_V, whatever the process, because it depends only on the change in temperature.

(b) Determine the work done on the gas. [2]

Answer

$Q = \Delta U + W_{\text{by gas}} \Longrightarrow W_{\text{by gas}} = -2.50 \times 10^4 - (-1.79 \times 10^4) = -7.15 \times 10^3$ J 1

Work done on the gas $= +7.15 \times 10^3$ J 1

[Check: $W_{by\ gas} = p\Delta V = nR\Delta T = (21.5)(8.31)(-40) = -7.15 \times 10^3$ J]

2

Explanation: The gas contracts as it cools at constant pressure, so the surroundings do work on the gas. In the form $Q = \Delta U + W$ the work is the work done **by** the gas, which is negative; the work done **on** the gas is the same size but positive.

(c) What will happen to the heat liberated by the gas if the cooling process is done at constant volume? Give your reason. [2]

Answer

The heat liberated is less (only 1.79×10^4 J)	1
because at constant volume no work is done, so the heat liberated equals only the decrease in internal energy	1
	2
Total for question	**7**

Explanation: For the same fall in temperature the decrease in internal energy is the same. At constant pressure the work done on the gas also has to be removed as heat, so more heat is liberated.

Section C [30 marks]

18. (a)

(i) A body moves with uniform speed in a circular path. Explain why the body experiences acceleration. [3]

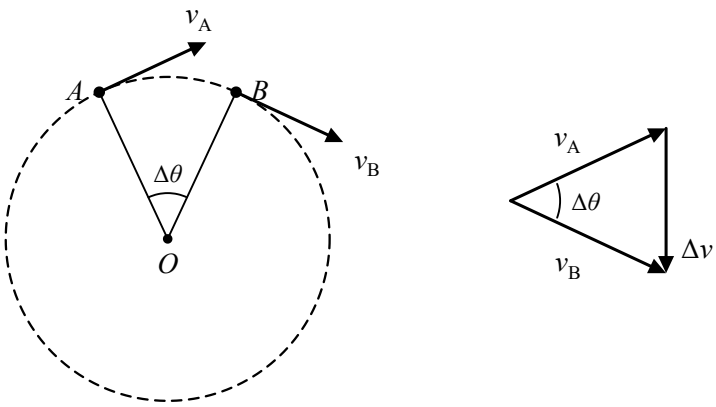
Answer

Velocity is a vector quantity	1
The direction of motion (velocity) changes continuously even though the speed is constant	1
A change in velocity means there is an acceleration (directed towards the centre)	1
	3

Explanation: Acceleration is the rate of change of velocity, not of speed. In circular motion the velocity is always along the tangent, so its direction keeps changing.

(ii) Derive an expression for the acceleration in terms of speed, v, and radius, r, of the circle. [5]

Answer



A body moves from A to B in time Δt, turning through a small angle $\Delta\theta$; $|\boldsymbol{v}_A| = |\boldsymbol{v}_B| = v$ 1

Change in velocity $\Delta\boldsymbol{v} = \boldsymbol{v}_B - \boldsymbol{v}_A$; from the vector triangle, for small $\Delta\theta$, $\Delta v = v\Delta\theta$ 1

$a = \dfrac{\Delta v}{\Delta t} = v\dfrac{\Delta\theta}{\Delta t}$ 1

As $\Delta t \to 0$, $\dfrac{\Delta\theta}{\Delta t} = \omega = \dfrac{v}{r}$ 1

$a = \dfrac{v^2}{r}$, directed towards the centre 1

5

Explanation: The vectors $\boldsymbol{v}_A$ and $\boldsymbol{v}_B$ are perpendicular to the radii OA and OB, so the angle between them is also $\Delta\theta$. For a small angle the vector triangle is a thin isosceles triangle whose short side is $v\Delta\theta$, and $\Delta\boldsymbol{v}$ points towards the centre.

(b) A steel ball of mass of 50.0 g moves with a constant speed in a horizontal circle of radius 2.0 m. The period of motion is 1.6 s. Calculate the centripetal force which acts on the mass. [3]

Answer

$v = \dfrac{2\pi r}{T} = \dfrac{2\pi(2.0)}{1.6} = 7.85 \text{ m s}^{-1}$ 1

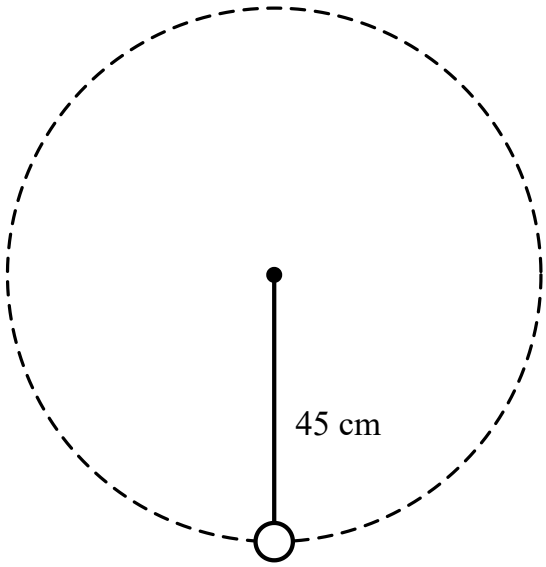
$F = \dfrac{mv^2}{r} = \dfrac{(0.0500)(7.85)^2}{2.0}$ 1

$F = 1.54$ N 1

3

Explanation: Find the speed (or $\omega = 2\pi/T$) from the period, then use $F = mv^2/r$ (or $mr\omega^2$). Convert 50.0 g to 0.0500 kg.

(c) An object is tied to a string of length 45 cm and whirled around in a vertical circle with constant speed as shown in the diagram below.



Determine the minimum speed for the object so that it can move around the vertical circle. [4]

Answer

At the top of the circle: $T + mg = \frac{mv^2}{r}$ 1

The speed is minimum when the tension $T = 0$ at the top 1

$mg = \frac{mv^2}{r} \Rightarrow v = \sqrt{(gr)} = \sqrt{((9.81)(0.45))}$ 1

$v = 2.10 \text{ m s}^{-1}$ 1

4

Total for question **15**

Explanation: The top of the circle is the critical point, because there the weight alone can supply the centripetal force. If the speed is less than $\sqrt{(gr)}$ the string goes slack and the object leaves the circle. Use the length in metres.

19. (a)

(i) State **two** assumptions of the kinetic theory of an ideal gas. [2]

Answer

• A gas consists of a very large number of molecules in continuous random motion 1

• The volume of the molecules is negligible compared with the volume of the container 1

• Collisions between molecules and with the walls are perfectly elastic 1

• There are no intermolecular forces except during collisions 1

[Any two – maximum 2 marks]

2

Explanation: Use the word molecules, not particles. Other acceptable assumptions: the duration of a collision is negligible compared with the time between collisions; the molecules obey Newton's laws of motion.

(ii) Show that the pressure exerted by an ideal gas could be determined using $p = \frac{1}{3}\rho\langle c^2\rangle$, where ρ and $\langle c^2\rangle$ are density and mean square speed of the gas respectively. [5]

Answer

Consider a molecule of mass m moving with velocity c_x in a cube of side L. On hitting a wall, change in momentum $= -mc_x - mc_x = -2mc_x$ — 1

Time between collisions with the same wall $= \frac{2L}{c_x}$, so force on the wall $F = \frac{2mc_x}{2L/c_x} = \frac{mc_x^2}{L}$ — 1

For N molecules: $p = \frac{F}{L^2} = \frac{m(c_{1x}^2 + c_{2x}^2 + \dots)}{L^3} = \frac{Nm\langle c_x^2\rangle}{V}$ — 1

Random motion: $\langle c_x^2\rangle = \langle c_y^2\rangle = \langle c_z^2\rangle = \frac{1}{3}\langle c^2\rangle \Rightarrow p = \frac{1}{3}\frac{Nm}{V}\langle c^2\rangle$ — 1

$\rho = \frac{Nm}{V} \Rightarrow p = \frac{1}{3}\rho\langle c^2\rangle$ — 1

5

Explanation: The derivation follows the chain: momentum change per collision → force from one molecule (rate of change of momentum) → pressure from all N molecules → average over the three directions → density.

(b) The relationship between pressure and molecular kinetic energy is given by $p = \frac{2}{3}(\frac{N}{V})(\frac{1}{2}m\langle c^2\rangle)$

(i) Calculate the average molecular kinetic energy for each helium and argon gas which is contained as a mixture in a cylinder in equilibrium at 150 °C. [3]

Answer

$\langle E_k\rangle = \frac{3}{2}kT$, with $T = 150 + 273 = 423$ K — 1

$\langle E_k\rangle = \frac{3}{2}(1.38 \times 10^{-23})(423) = 8.76 \times 10^{-21}$ J — 1

Both helium and argon have the same average kinetic energy, 8.76×10^{-21} J, because they are at the same temperature — 1

3

Explanation: From the given relation and $pV = NkT$, $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$. The average translational kinetic energy depends only on the temperature, not on the type of gas.

(ii) Determine the r.m.s. speed of each type of gas molecule in (b)(i) if the masses of helium and argon are 6.64×10^{-27} kg and 6.63×10^{-26} kg respectively. [3]

Answer

$c_{\text{r.m.s.}} = \sqrt{\left(\frac{2\langle E_k \rangle}{m}\right)}$ 1

Helium: $c_{\text{r.m.s.}} = \sqrt{\left(\frac{2(8.76 \times 10^{-21})}{6.64 \times 10^{-27}}\right)} = 1.62 \times 10^{3}$ m s^{-1} 1

Argon: $c_{\text{r.m.s.}} = \sqrt{\left(\frac{2(8.76 \times 10^{-21})}{6.63 \times 10^{-26}}\right)} = 514$ m s^{-1} 1

3

Explanation: With equal kinetic energies, the lighter helium molecules move faster: $c_{\text{r.m.s.}} \propto 1/\sqrt{(m)}$. Use the mass of one molecule, not the molar mass.

(c) Describe the degree of freedom in a gas. [2]

Answer

The degrees of freedom are the number of independent ways in which a molecule can possess (store) energy 1

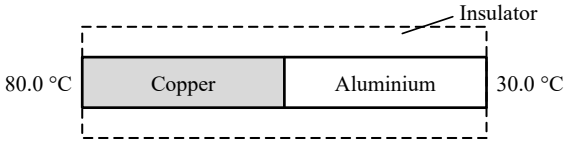
e.g. a monatomic molecule has 3 (translational); a diatomic molecule has 5 (3 translational + 2 rotational) at room temperature 1

2

Total for question **15**

Explanation: By the principle of equipartition of energy, each degree of freedom has an average energy of $\frac{1}{2}kT$ per molecule.

20. (a) Two rods, a copper and an aluminium, of equal length and cross-sectional area are joined together and perfectly insulated as shown in the diagram below.



The temperatures at the open ends of the copper and aluminium are 80.0 °C and 30.0 °C respectively.

[The thermal conductivity of the copper and aluminium are 401 W m^{-1} $°C^{-1}$ and 235 W m^{-1} $°C^{-1}$ respectively.]

(i) Describe the heat transfer mechanism in the copper rod. [3]

Answer

At the hot end the atoms vibrate with larger amplitude and the free electrons gain kinetic energy	1
The free electrons move (diffuse) quickly to the cooler end and transfer energy by colliding with the atoms there	1
Energy is also passed along by the vibrating atoms colliding with neighbouring atoms (lattice vibrations); in a metal the free-electron mechanism is dominant	1
	3

Explanation: Copper is a metal, so it has free electrons, which is why it conducts heat much better than a non-metal, where heat is passed only by lattice vibrations.

(ii) Calculate the temperature at the interface of the two rods. [3]

Answer

Steady state, same rate of heat flow: $\frac{dQ}{dt} = k_{Cu}A\frac{80.0-\theta}{L} = k_{Al}A\frac{\theta-30.0}{L}$	1
$401(80.0-\theta) = 235(\theta-30.0)$	1
$\theta = 61.5\ °C$	1
	3

Explanation: The rods are insulated and in series, so the same heat per second flows through both. Copper is the better conductor, so the temperature drop across it is smaller, and the interface temperature is closer to 80.0 °C.

(b) An isolated object with a surface area of 3.0 m^2, temperature of 25 °C and emissivity of 0.80 is placed in an environment of 75 °C for 50.0 s.

(i) State the factors which affect the amount of energy radiated by the object. [3]

Answer

• Temperature of the surface of the object	1
• Surface area of the object	1
• Nature of the surface (emissivity) of the object	1
[The time of exposure also affects the total energy]	
	3

Explanation: These factors come from Stefan–Boltzmann's law, $P = e\sigma AT^4$.

(ii) Determine the net rate of energy exchanged of the object. [4]

Answer

$T = 25 + 273 = 298$ K; $T_0 = 75 + 273 = 348$ K — 1

$P_{net} = e\sigma A(T_0^4 - T^4)$ — 1

$P_{net} = (0.80)(5.67 \times 10^{-8})(3.0)(348^4 - 298^4)$ — 1

$P_{net} = 923$ W (absorbed by the object) — 1

4

Explanation: The surroundings are hotter than the object, so the object absorbs more radiation than it emits. Use kelvin temperatures and raise each to the fourth power before subtracting.

(iii) Determine the net energy absorbed by the object. [2]

Answer

$E = P_{net}t = (923)(50.0)$ — 1

$E = 4.61 \times 10^4$ J (46.1 kJ) — 1

2

Total for question **15**

Explanation: Energy is power × time. This assumes the temperature of the object stays at about 25 °C during the 50.0 s.